

DUAL-CORE OPTICAL MOTION TESTBED

FIRMWARE, HOST SOFTWARE & RESULTS

STM32H747 MotionIPC, live telemetry, automated characterization, and measured operation

Prepared by	Herder Elektronische Systemen
Report scope	Dual-core firmware architecture, host GUI, telemetry integrity, test automation, and measured motion performance
Evidence basis	Source code, engineering plots, project photographs, CAD/PCB artifacts, and recorded demonstrations
Revision	4.0 - 5 August 2026

DOCUMENT PURPOSE

This report documents the software-controlled testbed: target-side motion control, dual-core interprocessor communication, operator software, data products, safeguards, and measured behavior. Hardware and PCB implementation are intentionally contained in the companion platform report.

1. System Role and Delivered Capability

The system combines a dual-core STM32H747 target with a host application that commands motion, displays live encoder and speed telemetry, executes repeatable tests, and writes analysis-ready records. The public evidence package separates control-plane messaging from high-rate motion-data movement and preserves the distinction between demonstrated behavior and recommended release actions.

- Cortex-M7 and Cortex-M4 workloads connected through OpenAMP/RPMsg and a custom AXI-SRAM shared ring.
- Live host console for connection, scripted test selection, telemetry plots, CSV logging, and metadata capture.
- Measured low-speed tracking, constant-speed error, step response, and closed-loop frequency-response evidence.
- Fault, heartbeat, sequence-gap, dropped-record, and ring-status observability.

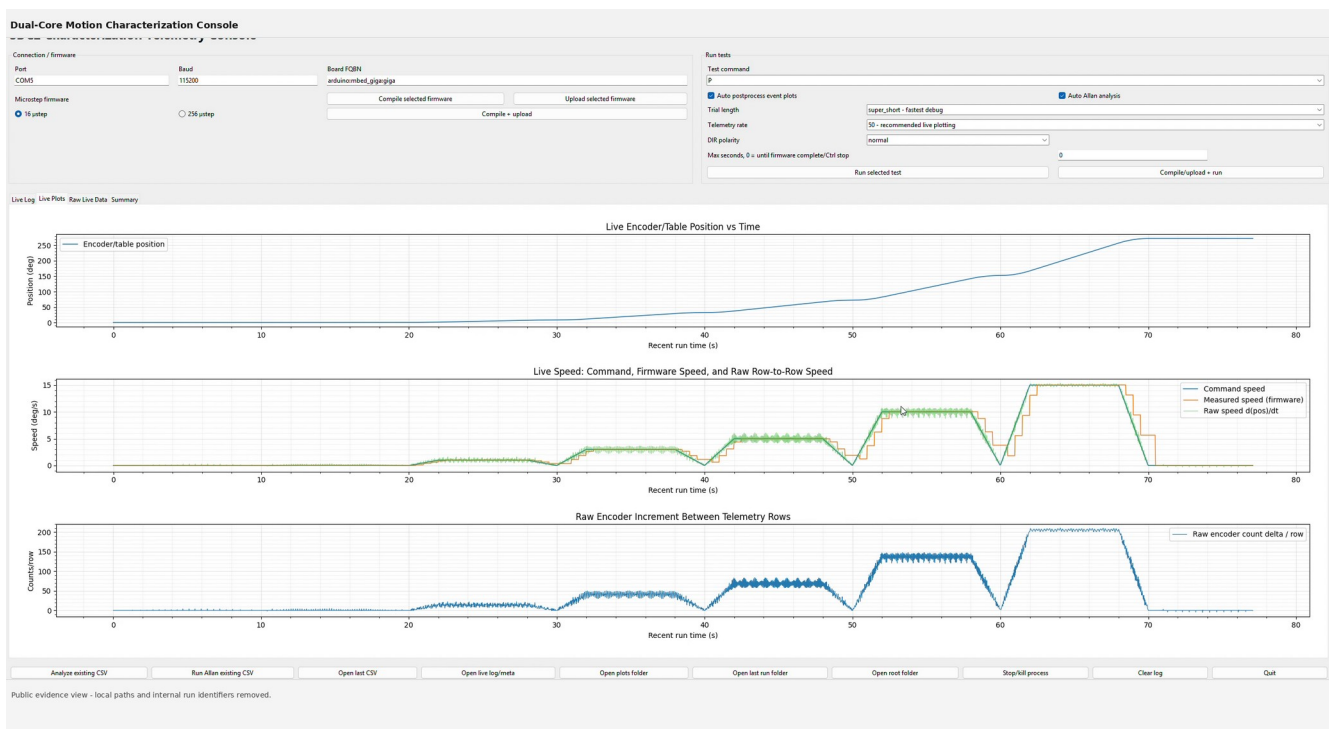


Figure 1. Live host telemetry console during an automated characterization sequence.

2. Dual-Core Firmware Architecture

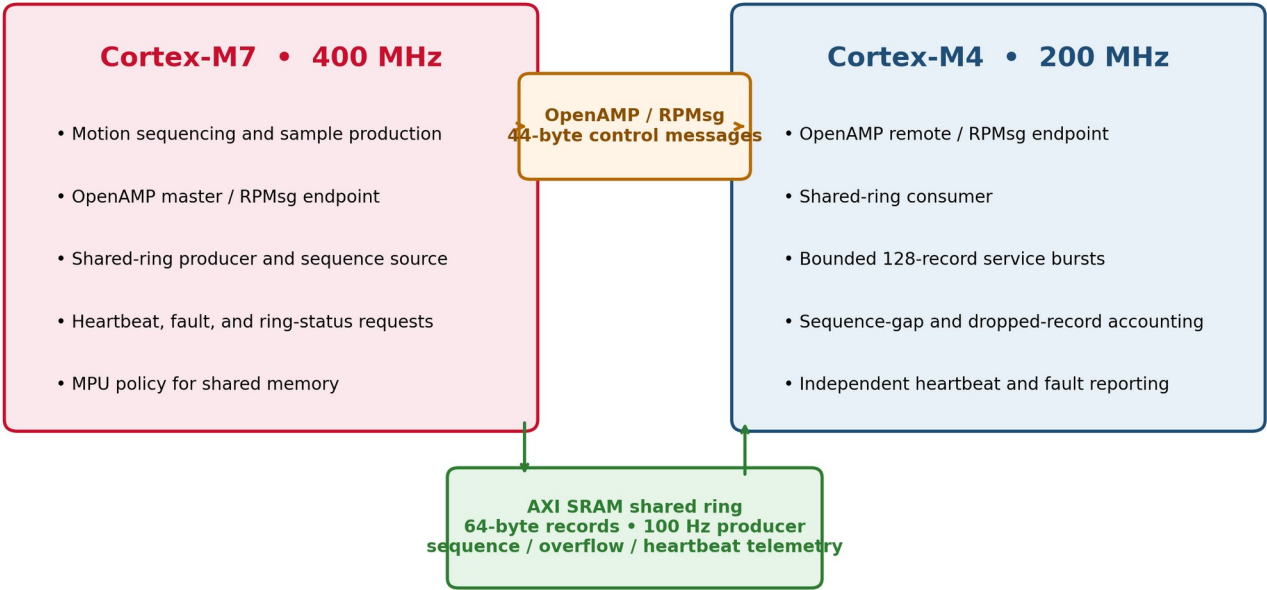


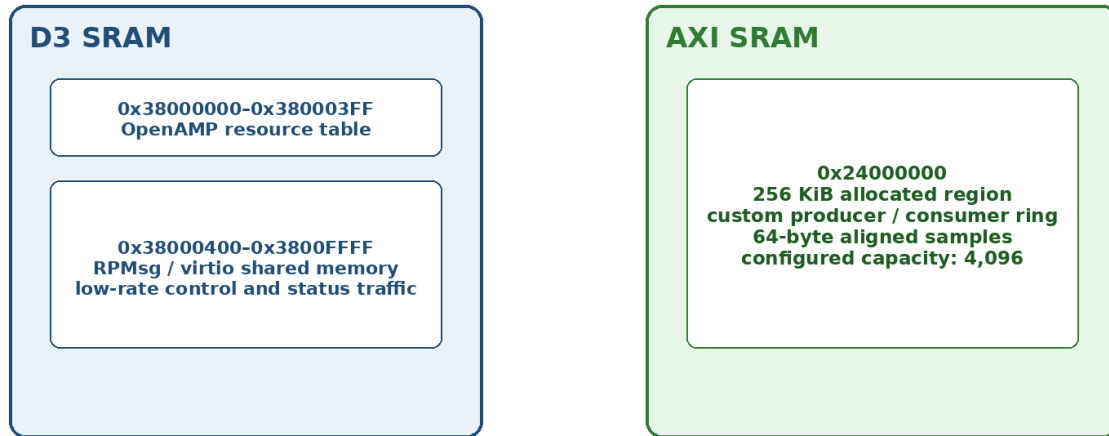
Figure 2. Dual-core architecture with physically separated control-message and shared-ring paths.

Interface	Verified role	Representative parameter
OpenAMP / RPMsg	Low-rate command and status coordination	44-byte application message
AXI-SRAM ring	High-rate aligned motion records	64-byte records; 4,096 nominal entries
M7 producer	Motion sequencing and sample publication	100 Hz sample production in the reviewed package
M4 consumer	Bounded service and accounting	Up to 128 records per service burst

3. Shared Memory, Timing, and Integrity

MotionIPC Shared-Memory Map

Separate regions preserve ST OpenAMP resources while providing a high-throughput custom data path



Engineering review finding: 4,096 × 64-byte records plus metadata exceed the 256 KiB allocation by ~64 b

Figure 3. MotionIPC shared-memory allocation and the engineering review finding on nominal ring capacity.

- OpenAMP resources occupy D3 SRAM while the custom producer/consumer ring is allocated in AXI SRAM.
- Records are aligned to cache-line boundaries and include sequence, overflow, and heartbeat information.
- Engineering review identified that 4,096 records plus metadata exceed a 256 KiB allocation by approximately one record; release should use 4,095 entries or enlarge the reserved region.
- The target reports independent heartbeat and fault state so host-side silence is distinguishable from target-side failure.

4. Host Application and Test Workflow

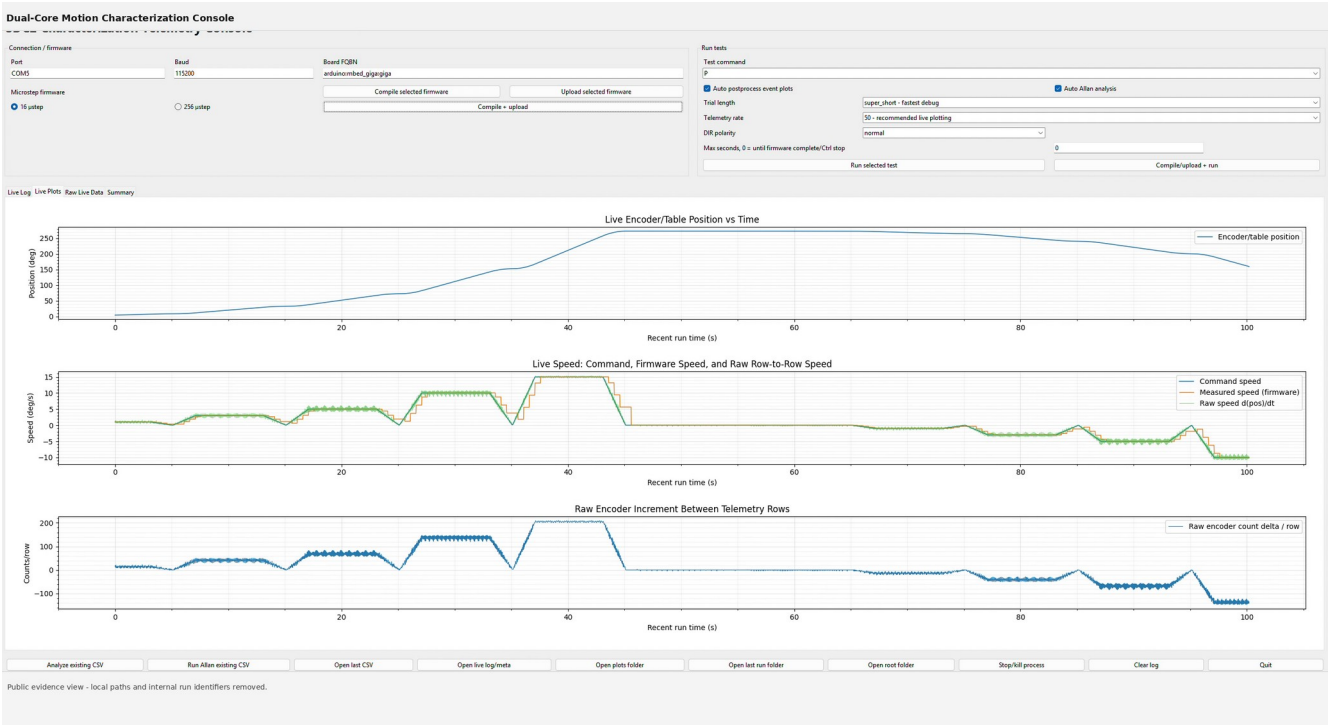


Figure 4. Host console view showing live position, commanded/measured speed, and raw encoder increments.

Workflow stage	Operator action	Recorded output
Connect	Select serial port and baud rate	Target identity and connection state
Configure	Select firmware and test definition	Run metadata and configuration
Execute	Start automated sequence	Live plots and telemetry rows
Review	Open generated outputs	CSV, figures, logs, and summary metrics

5. Low-Speed Tracking Results

SDC2 ONLY | 256 MICROSTEP BASELINE

Speed tracking summary: usable ultra-slow range, clear high-speed ceiling

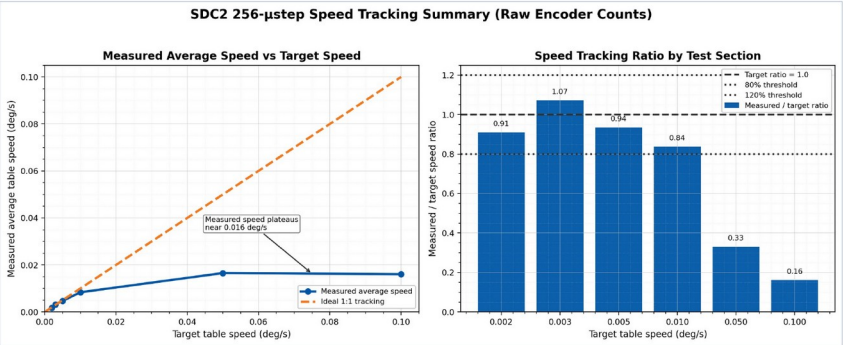


Figure 1. Clean summary view with legend moved off the data region. Average speed is computed from total raw encoder count delta over each section.

Main result

- 0.002-0.01 deg/s tracks in the correct range at the current 256 microstep setting.
- Best tracking occurs around 0.003-0.005 deg/s; 0.01 deg/s is usable but below target.
- 0.05 and 0.10 deg/s fail similarly and plateau near 0.016 deg/s actual speed.
- Interpretation: higher commanded speeds likely require lower microstepping and/or controller tuning.

Numerical baseline

- 0.002 -> 0.001818 deg/s, ratio 0.909
- 0.003 -> 0.003214 deg/s, ratio 1.071
- 0.005 -> 0.004676 deg/s, ratio 0.935
- 0.010 -> 0.008378 deg/s, ratio 0.838
- 0.050 -> 0.016560 deg/s, ratio 0.331
- 0.100 -> 0.016071 deg/s, ratio 0.161

SDC2 Encoder-Based Speed Tracking

Raw encoder-event analysis | no filtering or grouped averaging

1

Figure 5. Speed-tracking summary showing the usable ultra-slow range and high-speed ceiling.

Commanded speed (deg/s)	Measured speed (deg/s)	Interpretation
0.002	0.001818	Trackable near the lower practical limit
0.003	0.003214	Best agreement region
0.005	0.004676	Strong low-speed tracking
0.010	0.008378	Usable with increasing bias
0.050-0.100	~0.016 plateau	Configuration-limited ceiling

6. Constant-Speed and Step-Response Characterization

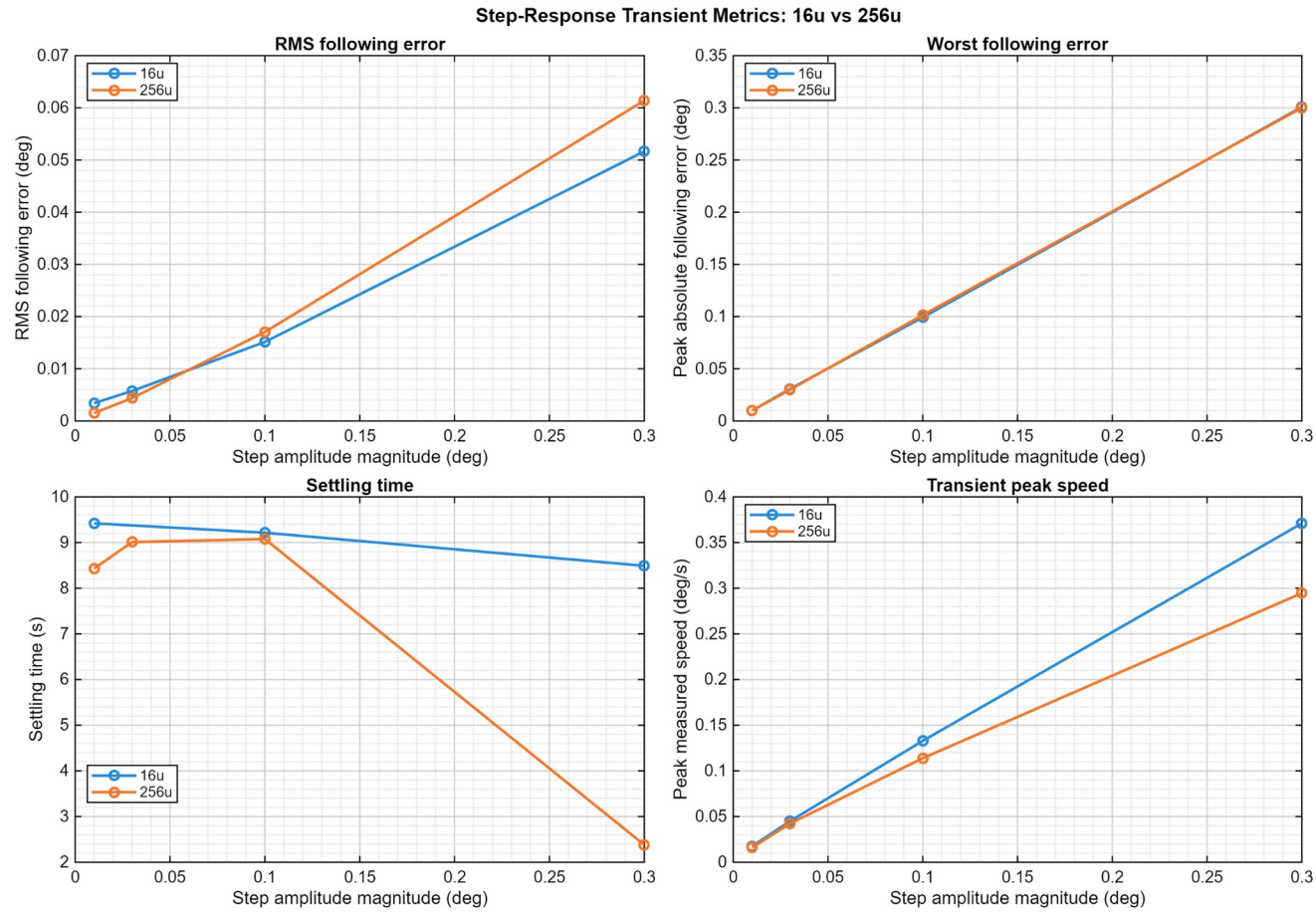


Figure 6. Extended constant-speed tracking and Allan-style stability summary.

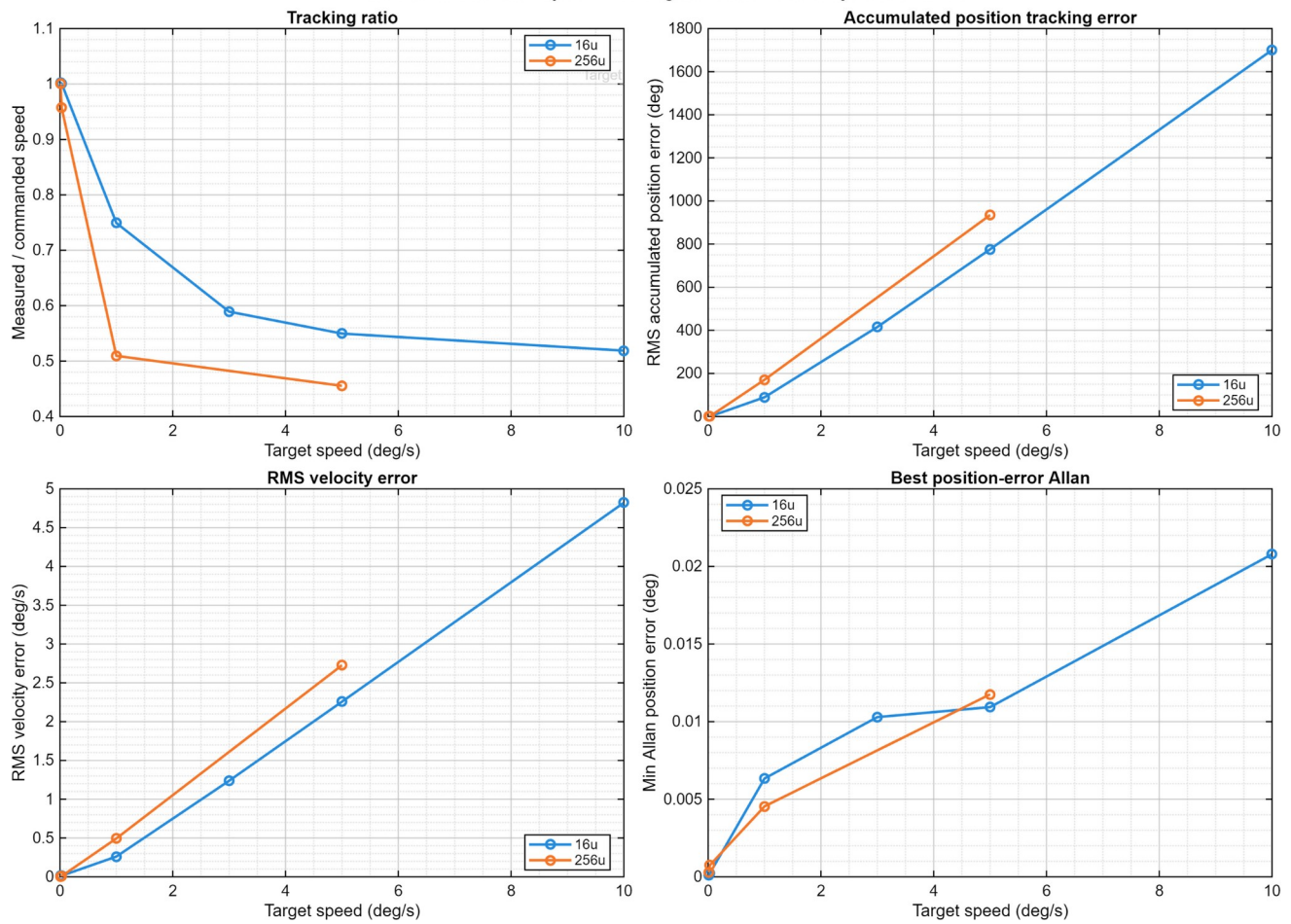


Figure 7. Step-response metrics for RMS following error, worst following error, gain, and position-area measures.

7. Closed-Loop Frequency Response

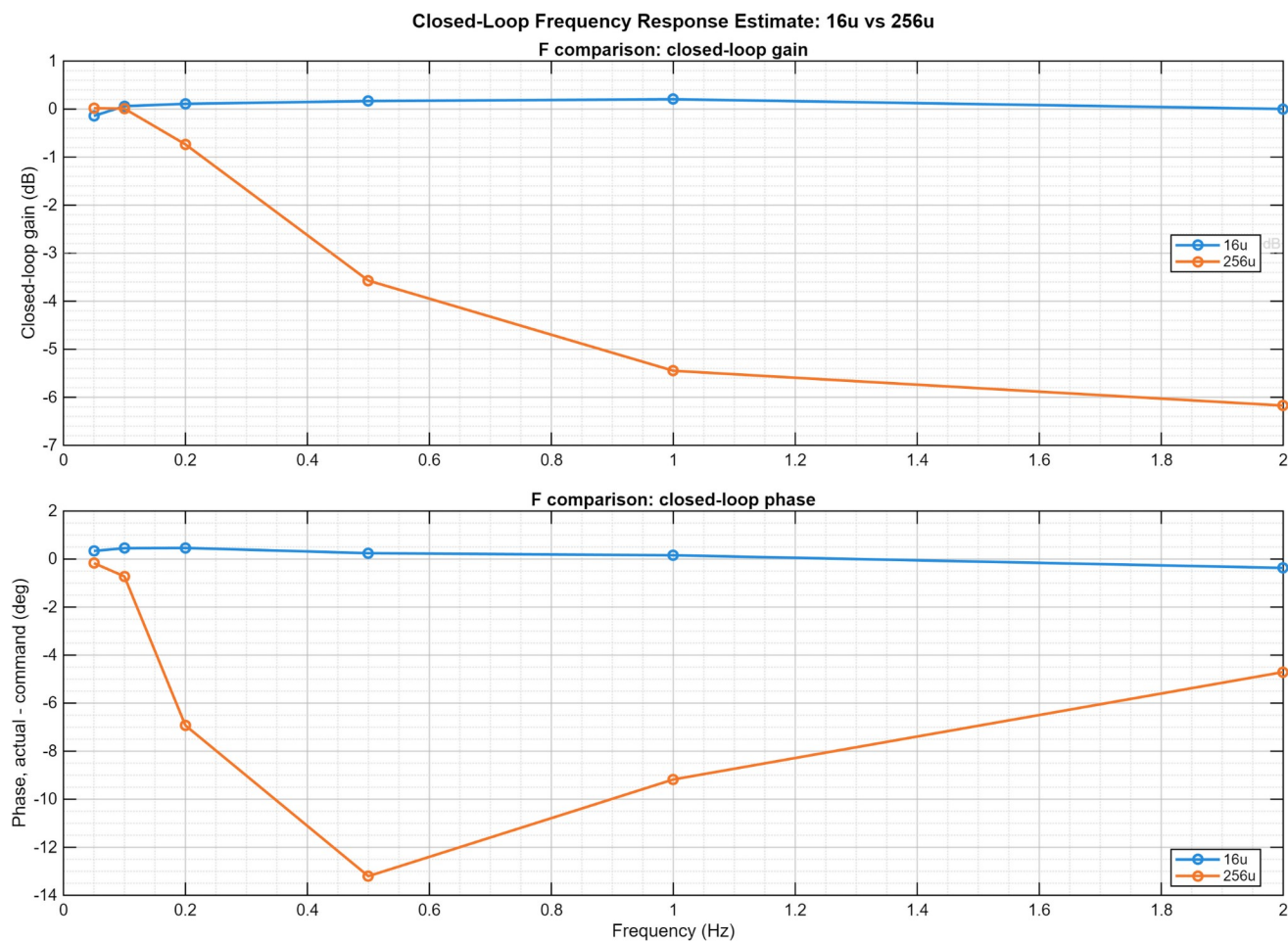


Figure 8. Closed-loop response estimate across the tested 0.33-2 Hz range.

The available evidence supports reporting response across the tested frequencies. It does not, by itself, establish a formal -3 dB bandwidth because the plotted data do not demonstrate a complete crossing with sufficient surrounding points.

8. Safety, Reliability, and Release Controls

Control	Implemented behavior	Release consideration
Host heartbeat	Periodic host presence message	Target deadman should remain enabled
Pulse-rate cap	25,000 pulses/s maximum	Verify against final drivetrain and driver
Backlog bound	Maximum 25 pulses emitted per loop	Prevents unlimited catch-up bursts
Telemetry limits	Selectable 10/20/50/100 Hz	Full ASCII rows may constrain sustainable rate
Fault accounting	Sequence gaps, drops, heartbeat, fault counters	Expose in final operator status panel

At 115,200 baud, full 24-field ASCII telemetry rows cannot be assumed to sustain 100 Hz under all conditions. The configured option should be treated separately from demonstrated end-to-end serial throughput; 50-60 Hz is a more realistic full-row operating range without a binary protocol or higher link rate.

9. Limitations and Document Control

Item	Controlled statement
Prototype maturity	No formal shock, vibration, thermal, EMC, or long-duration certification is claimed.
MotionIPC package	Reviewed v0.1 sample producer is 100 Hz and should not be represented as the final integrated motor-control workload without the corresponding integrated revision.
Bandwidth claim	Tested response is reported; formal bandwidth is not claimed.
Release package	Preserve tagged source, host build, test definitions, and representative datasets together.

Attribution

Herder Elektronische Systemen developed the public-facing engineering package and documents the dual-core firmware integration, host software, test workflows, and measured evidence represented here. No third-party affiliation or endorsement is claimed.

Revision record

Revision	Date	Description
4.0	2026-08-05	Rebuilt from scratch with restored GUI, architecture, and measured-result imagery; removed inherited layout defects and preserved evidence boundaries.